

Sum The File

The Process:

For this code, I built it up one thing at a time. I first started with making code that could open and read a file with the filename as a given string and printed out the file contents.

I then worked on a string with several single digit numbers separated by the newline character. This was for easier testing while making the code. With this, I traversed through the string and checked each character for if it was a 0 or a newline. If it was a newline, I knew that I got a number and could add it to the total. If it was a 0, that means I'd gotten to the end and the final number should be added, then it can be ended.

At this point, I needed to be able to print multiple digit numbers. I came up with a system where I use the stack to make sure the digits were printed out in the correct order. I used the div operation to get the remainder and pushed that remainder onto the stack. Once the quotient was 0, I knew to stop dividing. Since the stack works as a first in last out, the last remainder would be the first thing pushed back out. I would then print the stack, using ecx as a digit counter to make sure I don't push the stack too much.

I then worked on checking for several digit numbers. I went about this by multiplying any previous numbers by 10, then adding the current number to that multiplied number. When a newline is reached, I knew that the number is done and can be added to the total.

I then worked to include the fact that the first number will be the number of integers in the file. I made it so that ebp would act a counter for how many numbers would be allowed to be read. For reading the first number, I set ebp to 1 and once the first number was read, doneCounting would be called to return to _start. I then set ebp to that first number. This acts as the counter so when the specified number of integers are read or 0 is read, the final total will be added and printed.

At this point, I reintegrated reading from the file. Finally, I worked on getting the user to say what file they want to read from. I added a loop to find and get rid of the newline character that happens when the user enters the filename.

The Code:

section .data

```
user_file_name_text db "Please enter a file name: ", 0
len_user_file_name_text equ $ - user_file_name_text
new_line db "", 0xA
len_new_line equ $ - new_line
```

section .bss

```
file_name resb 100
file_info resb 6006
current_num resd 1
current_digit resb 1
total resd 1
```

```

        print_num resb 1

section .text
    global _start

_start:
    call getFile
    call resetSetup

    mov ecx, file_name
    call removeNewLine
    call resetSetup

    call fileOpening
    call resetSetup

    mov ecx, file_info

    mov ebp, 1
    call sumTheFile

    mov ebp, esi
    xor esi, esi

    call sumTheFile
    call finalAdd

getFile:
    mov eax, 4
    mov ebx, 1
    mov ecx, user_file_name_text
    mov edx, len_user_file_name_text
    int 0x80

    mov eax, 3
    mov ebx, 2
    mov ecx, file_name
    mov edx, 100
    int 0x80
    ret

removeNewLine:
    mov bl, [ecx]

```

```
cmp bl, 0Ah
je foundNewLine
```

```
cmp bl, 0
je doneCounting
```

```
inc ecx
jmp removeNewLine
```

```
foundNewLine:
    mov byte [ecx], 0
    ret
```

```
fileOpening:
    mov eax, 5
    mov ebx, file_name
    mov ecx, 0
    int 0x80
```

```
    mov ebx, eax
    mov eax, 3
    mov ecx, file_info
    mov edx, 6006
    int 0x80
```

```
    mov eax, 6
    int 0x80
```

```
    ret
```

```
sumTheFile:
    mov bl, [ecx]

    cmp ebp, 0
    jle doneCounting

    cmp bl, 0
    je finalAdd

    cmp bl, 0Ah
    je newLineAdd

    sub bl, '0'
    mov [current_digit], bl
```

```
mov eax, 10
xor edx, edx
mul edi
```

```
mov [current_num], eax
mov edi, [current_num]
mov edx, [current_digit]
```

```
add edi, edx
```

```
inc ecx
jmp sumTheFile
```

newLineAdd:

```
add esi, edi
```

```
xor edi, edi
mov [current_num], edi
```

```
inc ecx
dec ebp
jmp sumTheFile
```

doneCounting:

```
ret
```

finalAdd:

```
add esi, edi
```

```
xor edi, edi
mov [current_num], edi
```

```
mov [total], esi
call resetSetup
```

getDigits:

```
xor edx, edx
mov ebx, 10
mov eax, [total]
div ebx
```

```
mov [total], eax
```

```
add edx, '0'
mov [print_num], edx
```

```
mov rsi, [print_num]
push rsi
```

```
cmp eax, 0
je startPrinting
```

```
inc ecx
jmp getDigits
```

startPrinting:

```
mov edi, ecx
```

printLoop:

```
pop rsi
mov [print_num], esi
```

```
mov eax, 4
mov ebx, 1
mov ecx, print_num
mov edx, 1
int 0x80
```

```
cmp edi, 0
je exit
```

```
dec edi
jmp printLoop
```

exit:

```
mov eax, 4
mov ebx, 1
mov ecx, new_line
mov edx, len_new_line
int 0x80
```

```
mov eax, 1
int 0x80
```

resetSetup:

```
xor eax, eax
xor ebx, ebx
```

```
xor ecx, ecx
xor edx, edx
xor esi, esi
xor edi, edi
ret
```

The Output:

With randomInt100.txt:

```
1. Duo Push to XXX-XXX-4197
2. Duo Push to iOS
3. Phone call to XXX-XXX-4197
4. SMS passcodes to XXX-XXX-4197

Passcode or option (1-4): 1
Success. Logging you in...
Success. Logging you in...

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If you have any questions or problems regarding these systems, please call the
DoIT Technology Support Center at 410-455-3838, or submit your request on the
web by visiting http://my.umbc.edu/help/request

Remember that the Division of Information Technology will never ask for your
password. Do NOT give out this information under any circumstances.
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Last login: Fri Mar 28 09:42:20 2025 from 130.85.248.49
[rmekash1@linux6 ~]$ cd cmpe310
[rmekash1@linux6 ~/cmpe310]$ cd proj2
[rmekash1@linux6 proj2]$ ls
a.out file_summer.asm file_summer.o nums.txt randomInt100.txt
[rmekash1@linux6 proj2]$ nasm -f elf64 file_summer.asm
[rmekash1@linux6 proj2]$ ld file_summer.o
[rmekash1@linux6 proj2]$ ./a.out
Please enter a file name: randomInt100.txt
4579
```

With nums.txt:

```

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[rmekash1@linux6 proj2]$ ld file_summer.o
[rmekash1@linux6 proj2]$ ./a.out
Please enter a file name: randomInt100.txt
4579
[rmekash1@linux6 proj2]$ ./a.out
Please enter a file name: nums.txt
30101

```

Here is the nums.txt file:

```

File Edit Options Buffers Tools Text Help
10
1000
4000
1000
1100
2000
3000
5000
6000
7000
1

```